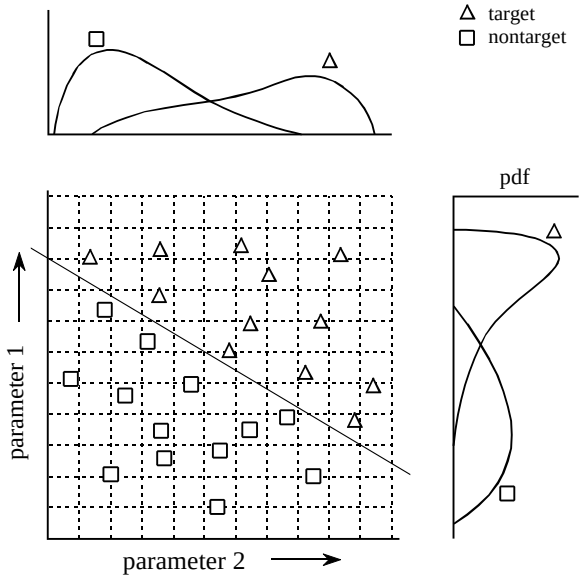

FIGURE 26-3
Example of a two-parameter space. The target (▵) and nontarget (□) groups are completely separate in two-dimensions; however, they overlap in each individual parameter. This overlap is shown by the one-dimensional pdfs along each of the parameter axes.

situation that makes you feel even worse: you speak with a patient who is terminally ill with a cancer that *your* system failed to detect. You respond to this difficult experience by *greatly* lowering the threshold. As time goes on and these events are repeated many times, the threshold gradually moves to an *equilibrium* value. That is, the false-positive rate multiplied by a significance factor (lowering the threshold) is balanced by the false-negative rate multiplied by another significance factor (raising the threshold).

This analysis can be extended to devices that provide more than one output. For example, suppose that a cancer detection system operates by taking an x-ray image of the subject, followed by automated image analysis algorithms to identify tumors. The algorithms identify suspicious regions, and then measure key characteristics to aid in the evaluation. For instance, suppose we measure the *diameter* of the suspect region (parameter 1) and its *brightness* in the image (parameter 2). Further suppose that our research indicates that tumors are generally larger and brighter than normal tissue. As a first try, we could go through the previously presented ROC analysis for each parameter, and find an acceptable threshold for each. We could then classify a test as positive only if it met both criteria: parameter 1 greater than some threshold *and* parameter 2 greater than another threshold.

This technique of thresholding the parameters separately and then invoking logic functions (AND, OR, etc.) is very common. Nevertheless, it is very inefficient, and much better methods are available. Figure 26-3 shows why this is the case. In this figure, each triangle represents a single occurrence of a target (a patient with cancer), plotted at a location that corresponds to the value of its two parameters. Likewise, each square represents a single occurrence of a nontarget (a patient without cancer). As shown in the pdf